Answer Key & Teacher Notes

Session 1 — Writing Numerical Expressions · Session 2 — Order of Operations with Powers

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Writing Numerical Expressions

TALK PROMPTS

- How do you decide whether a story needs you to add, to multiply, or to do both?
- In Tickets and Programs, why do the two products get added instead of multiplied together? Point to the words that tell you.
- Why does a numerical expression have no equal sign?

PARTNER TASK

Field Trip and Tickets. One partner writes $20 \times 13 + 25$ for \$13 per student and a \$25 bus fee; the other evaluates it (\$285). Switch roles with 5 tickets at \$8 and 4 programs at \$5 ($5 \times 8 + 4 \times 5 = \$60$).

WATCH FOR

- Adding every number in the story instead of multiplying equal groups ($3 + 16 - 5$ for 3 bags of 16 crackers, minus 5 eaten).
- Writing "15 less than 40" as $15 - 40$.
- Leaving out parentheses when a whole sum or total must be found first ($100 - 3 \times 12 + 2 \times 20$).

SESSION 2 · TEACHER NOTES

Order of Operations with Powers

TALK PROMPTS

- Why do mathematicians agree on one order instead of letting each person choose?
- In $40 - (6 + 2)^2 \div 4$, the division comes before the subtraction. Which rule puts it there?

PARTNER TASK

Same Numbers, Two Values. One partner evaluates $20 - 3 \times 4$ (8); the other evaluates $(20 - 3) \times 4$ (68). Then switch roles with $40 - (6 + 2)^2 \div 4$ (24) and $7 \times 2^3 - 9$ (47), naming each step aloud.

WATCH FOR

- Working strictly left to right and adding before multiplying ($8 + 4 \times 3^2$ as 12×9).
- Applying an exponent to a product (2×5^2 as 10^2).
- Thinking multiplication always comes before division, or addition before subtraction.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.EE.A.2	1	2	1	1	2	2	3	2	2	1	1
S1 · Form B	6.EE.A.2	1	2	1	1	2	2	3	2	2	1	1
S2 · Form A	6.EE.A.2c	1	2	1	1	2	3	2	2	2	1	1
S2 · Form B	6.EE.A.2c	1	2	1	1	2	3	2	2	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

numerical expression: Numbers and operations with no **equal** sign.

evaluate: Find the one **number** an expression equals.

product: The answer when you **multiply**.

- Equal groups ("each," "per") in a story usually mean **multiply**.
- A total of different groups means **add** the products.
- "Less than" flips the order: 5 less than 12 is **12** − 5.

Example: Shirts are equal groups: 6 × **7**. Add the one hat: 6 × 7 + **5**. Evaluate: 42 + 5 = **47**. Answer: 6 × 7 + 5 = $**47**

YOU TRY

1. a. 6 × 4 + 2 b. 35 − 10

2. 3 × 5 + 4 = 15 + 4 = $19

EXIT TICKET

1. a. 30 − 9 b. 3 × (4 + 5)

2. 6 × 2 + 3 = 12 + 3 = $15

3. **A** (4 × 7 − 5)

PRACTICE FORM A

1. a. 6 × 3 + 4 b. 75 − 20 c. 3 × (5 + 9) d. 48 ÷ 8 + 7

2. a. 5 × 12 + 4 × 3 = 60 + 12 = $72 b. 10^2 = 100 notebooks

3. 20 × 13 + 25 = 260 + 25 = $285

4. 15^2 = 15 × 15 = 225 plants

5. Leo is wrong. "Each" means equal groups, so multiply: 3 × 16 − 5 = 48 − 5 = 43 crackers.

6. 12^3 = 12 × 12 × 12 = 1,728 cubes

7. a. 3 × 12 + 2 × 20 (= $76) b. 100 − (3 × 12 + 2 × 20) = 100 − 76 = $24 c. The parentheses show that the whole cost is found first, then subtracted from $100.

8. 4 × 9 + 3 × 6 = 36 + 18 = $54

9. The 4 snacks at $3 each belong in a product: 3 × 4. The poster is added once. Correct: 3 × 4 + 10 = 12 + 10 = $22.

10. **A** (5 × 4 + 3)

11. **B** (40 − 15)

Self-check word **PRODUCT**

PRACTICE FORM B

1. a. 7 × 2 + 6 b. 60 − 18 c. 2 × (3 + 7) d. 56 ÷ 7 − 3

2. a. 6 × 11 + 3 × 8 = 66 + 24 = $90 b. 9^2 = 81 tiles

3. 25 × 12 + 30 = 300 + 30 = $330

4. 14^2 = 14 × 14 = 196 plants

5. Mia is wrong. "Each" means equal groups, so multiply: 4 × 15 − 6 = 60 − 6 = 54 pretzels.

6. 10^3 = 10 × 10 × 10 = 1,000 cubes

7. a. 4 × 10 + 3 × 15 (= $85) b. 100 − (4 × 10 + 3 × 15) = 100 − 85 = $15 c. The parentheses show that the whole cost is found first, then subtracted from $100.

8. 5 × 8 + 4 × 5 = 40 + 20 = $60

9. 6 trays of 8 is equal groups, so 6 and 8 belong in a product. Correct: 6 × 8 + 9 = 48 + 9 = 57 seedlings.

10. **B** (2 × (8 + 2))

11. **A** (do first)

Self-check word **NUMBERS**

GUIDED NOTES

order of operations: The rules for which **step** to do first, next and last.

grouping symbols: Parentheses or brackets; do the work **inside** them first.

left to right: Do partner steps (× and ÷, or + and −) in the order they **appear**.

- First do work inside **grouping** symbols, then evaluate **powers**.
- Next multiply and divide from **left** to right.
- Last, add and subtract from left to **right**.

Example: Parentheses: 2 + 3 = **5**, so 40 − 5^2 ÷ 5. Power: 5^2 = **25**, so 40 − 25 ÷ 5. Divide, then subtract: 40 − **5**. Answer: 40 − $(2 + 3)^2$ ÷ 5 = **35**

YOU TRY

1. a. 2 b. 16

2. 2^3 = 8; 8 × 3 = 24; 6 + 24 = 30

EXIT TICKET

1. a. 14 b. 30

2. 2^2 = 4; 4 × 3 = 12; 20 − 12 = 8

3. **A** (32)

PRACTICE FORM A

1. a. 8 b. 68 c. 13 d. 54 e. 12 f. 11

2. a. 3^2 = 9 b. 4 × 9 = 36 c. 8 + 36 = 44; 44 − 10 = 34 d. Left to right adds 8 + 4 first: 12 × 9 − 10 = 98. Multiplication must come before addition.

3. 6^2 ÷ 4 + 13 = 36 ÷ 4 + 13 = 9 + 13 = 22

4. 5^2 + 8 × 6 ÷ 2 = 25 + 48 ÷ 2 = 25 + 24 = 49

5. Dara added 2 + 3 first. Correct: 4^2 = 16, 3 × 16 = 48, 2 + 48 = 50.

6. Sample: (24 ÷ 4 + 2) × 3 = 8 × 3 = 24, or 24 ÷ (4 + 2 × 3) = 24 ÷ 10 = 2.4. (Note: 24 ÷ (4 + 2) × 3 is still 12.)

7. Kwame multiplied 4 × 3 before squaring. Correct: 3^2 = 9, 4 × 9 = 36, 36 + 6 = 42.

8. a. 4 + 12 = 16 b. 6 × 6 = 36 c. Without parentheses, multiply first; the parentheses make you add first.

9. Powers come before multiplication, and the exponent applies only to 5. Correct: 5^2 = 25, 2 × 25 = 50, 10 + 50 = 60.

10. **B** (20)

11. **A** (before × and ÷)

Self-check word **POWERS**

PRACTICE FORM B

1. a. 10 b. 130 c. 21 d. 24 e. 20 f. 17

2. a. 4^2 = 16 b. 3 × 16 = 48 c. 7 + 48 = 55; 55 − 12 = 43 d. Elena adds 7 + 3 first: 10 × 16 − 12 = 148. Multiplication must come before addition.

3. 6^2 ÷ 4 + 10 = 36 ÷ 4 + 10 = 9 + 10 = 19

4. 6^2 + 9 × 6 ÷ 3 = 36 + 54 ÷ 3 = 36 + 18 = 54

5. Theo added 3 + 2 first. Correct: 5^2 = 25, 2 × 25 = 50, 3 + 50 = 53.

6. Sample: (40 ÷ 5 + 3) × 2 = 11 × 2 = 22, or 40 ÷ (5 + 3) × 2 = 5 × 2 = 10.

7. Rosa multiplied 2 × 3 before cubing. Correct: 3^3 = 27, 2 × 27 = 54, 54 + 4 = 58.

8. a. 5 + 21 = 26 b. 8 × 7 = 56 c. Without parentheses, multiply first; the parentheses make you add first.

9. Powers come before multiplication, and the exponent applies only to 4. Correct: 4^2 = 16, 3 × 16 = 48, 9 + 48 = 57.

10. **B** (14)

11. **B** (2)

Self-check word **ORDERS**

SHEET 1 · WRITING NUMERICAL EXPRESSIONS

1. a. 7 × 8 − 10 b. 50 − 12 c. 36 ÷ 9 + 2 d. 2 × (15 − 6) e. 3 × 5 + 9 f. 3 × (42 ÷ 7)
2. 5 × 6 + 4 × 3 = 30 + 12 = $42
3. 12 × 2 + 8 × 5 − 15 = 24 + 40 − 15 = $49
4. (8 + 2) × 5
5. Sample: 4 friends each buy a $6 sandwich and a $3 drink. Total cost 4 × 9 = $36.

SHEET 2 · ORDER OF OPERATIONS WITH POWERS

1. a. 26 b. 50 c. 17 d. 6 e. 20 f. 24
2. a. (2 + 3) × 4 = 20 b. 24 ÷ (4 + 2) = 4
3. 3 × (64 − 4) = 3 × 60 = $180
4. 5 × 16 − 30 = 80 − 30 = 50 beads
5. Lou: add and subtract left to right, 18 − 6 = 12, 12 + 2 = 14 (Kim added first).
6. **A** (17)
7. **B** (2 + 3 × 4)

October 2026 conceptual additions — teacher key

Session 1 · Writing Numerical Expressions

4 × 6 + 3 = 27 matches. 4 × (6 + 3) = 36 would describe 3 additional markers in every box, not three singles.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: 4 × 6 + 3 = 27 matches. 4 × (6 + 3) = 36 would describe 3 additional markers in every box, not three singles.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Order of Operations with Powers

2 × 9 = 18; 6^2 = 36. The grouped expression is the base in the second. 24 ÷ 4 × 2 = 6 × 2 = 12, evaluated left to right.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: 2 × 9 = 18; 6^2 = 36. The grouped expression is the base in the second. 24 ÷ 4 × 2 = 6 × 2 = 12, evaluated left to right.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.